

Responsibilities:

- Design and implement autonomous agents using LangChain / Crew.ai / SK / Autogen and other relevant frameworks
- Develop and optimize RAG systems for improved information retrieval and response generation
- Create robust evaluation frameworks for testing agent performance and safety
- Implement monitoring and logging systems for agent behavior
- Develop documentation and best practices for agent development
- Contribute to the improvement of agent architecture and capabilities

Hands-on experience with:

- Building and deploying autonomous agents
- LangChain framework for developing applications
- Retrieval Augmented Generation (RAG) systems
- Large Language Models (LLMs) and their APIs
- Solid understanding of safety principles and best practices
- Resolving hallucinations in the outcomes

Mandatory Skill sets:

- Advanced Python programming with expertise in async programming and API development
- Experience with vector databases and embedding models
- Proficiency in implementing RAG pipelines and knowledge retrieval systems
- Familiarity with prompt engineering and LLM optimization techniques
- Experience with development frameworks and tools
- Knowledge of modern software development practices (Git, CI/CD, testing)
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Preferred Skill sets:

- Multiple LLM providers (OpenAI, Anthropic, etc.)
- Tools for agent orchestration and workflow management

- Natural Language Processing (NLP) techniques
- Container technologies (Docker, Kubernetes)
- Understanding of semantic search and information retrieval concepts
- Knowledge of agent alignment and safety considerations

***Responsibilities:**

- Design, develop, and implement generative AI models (e.g., GANs, VAEs, Transformers) for various applications such as text, image, audio, or video generation.
- Build scalable Python pipelines for training, fine-tuning, and deploying large language models and other generative architectures.
- Collaborate with data scientists, ML engineers, and product teams to integrate AI models into products and platforms.
- Optimize model performance, including speed, accuracy, and resource consumption.
- Conduct experiments to improve model robustness and creativity.
- Stay up-to-date with the latest research in generative AI and incorporate new techniques into our stack.
- Write clean, maintainable code and maintain proper documentation.
- Mentor junior engineers and contribute to code reviews.

***Mandatory skill sets**

- Strong proficiency in Python programming and related AI/ML libraries (TensorFlow, PyTorch, Hugging Face Transformers, etc.).
- Solid understanding of generative models such as GANs, VAEs, autoregressive models, and diffusion models.
- Experience with training and fine-tuning large language models.
- Familiarity with cloud platforms (AWS, GCP, Azure) and containerization tools like Docker/Kubernetes.
- Knowledge of data preprocessing, augmentation, and handling large datasets.
- Experience implementing end-to-end AI model deployment pipelines.
- Ability to write efficient, optimized code and profile AI workloads.

***Preferred skill sets**

- Experience with prompt engineering and working with large-scale LLMs (e.g., GPT, T5, BERT).
- Knowledge of reinforcement learning or unsupervised learning techniques.
- Familiarity with MLOps and CI/CD tools for model deployment.
- Publications or contributions to open source AI projects.
- Strong problem-solving skills and ability to work independently or as part of a team.

GenAI Application Development

- Develop GenAI applications using LLM APIs, prompt templates, structured outputs, RAG pipelines, embeddings, vector search, and tool calling.
- Build conversational AI, enterprise search, document Q&A, summarization, classification, data extraction, and workflow automation use cases.
- Integrate GenAI features with backend applications, portals, APIs, databases, document stores, and enterprise systems.
- Implement prompt orchestration, context management, response formatting, source citation handling, and feedback capture.

RAG, Semantic Search & Knowledge Ingestion

- Build document ingestion pipelines including parsing, OCR coordination, chunking, metadata extraction, embedding generation, indexing, and refresh.
- Implement semantic search and hybrid search using vector databases and search platforms.
- Support retrieval optimization through metadata filtering, chunking strategy, re-ranking, grounding, and citation generation.
- Work with enterprise knowledge sources such as SharePoint, Confluence, Google Drive, S3, databases, CRM, ITSM tools, and document repositories.

Agentic AI Development

- Develop basic to intermediate agentic workflows using LangChain, LangGraph, CrewAI, LlamaIndex, or equivalent frameworks.
- Implement agents with tools, memory, reasoning steps, API actions, and human-in-the-loop checkpoints.

- Build semi-autonomous workflows for business process automation while following safety and approval guardrails.

Backend, API & Data Engineering

- Develop backend services using Python, FastAPI, Node.js, or similar technologies.
- Build REST APIs for LLM orchestration, retrieval, ingestion, evaluation, semantic search, and agent execution.
- Work with SQL and NoSQL databases such as PostgreSQL, MySQL, SQL Server, MongoDB, Cosmos DB, DynamoDB, or equivalent.
- Use Redis or equivalent caching for session state, conversational memory, frequently accessed retrieval results, rate limiting, and workflow state.
- Follow secure coding, logging, exception handling, request validation, OpenAPI documentation, and production deployment practices.

Testing, Evaluation & Observability

- Support prompt testing, LLM response validation, RAG evaluation, regression testing, and hallucination checks.
- Use tools such as LangSmith, Langfuse, RAGAS, DeepEval, TruLens, Promptfoo, or equivalent under guidance.
- Capture logs, traces, token usage, latency, cost, feedback, and quality signals for GenAI applications.

Mandatory skill sets:

CORE GENAI - LLM APIs, prompt engineering, embeddings, RAG, semantic search, vector databases, structured outputs, grounding (Hands-on).

Frameworks - LangChain, LangGraph, CrewAI, LlamaIndex, Langflow (Hands-on).

Backend APIs - Python, FastAPI, Node.js, REST APIs, OpenAPI, async processing (Hands-on).

Data Stores - SQL, NoSQL, vector DB, Redis caching, object storage (Working knowledge).

Cloud AI - Azure OpenAI, AWS Bedrock, GCP Vertex AI, OpenAI, Anthropic Claude, Gemini, Hugging Face (Working knowledge).

Document Processing - PDF, Word, Excel, HTML, OCR, chunking, metadata extraction, indexing (Working knowledge).

Evaluation & Observability - LangSmith, Langfuse, RAGAS, Promptfoo, OpenTelemetry basics (Basic to working knowledge).

Security & Responsible AI - PII handling, access control, prompt injection awareness, content filtering, audit logging (Basic knowledge).

Soft Skills

- Strong problem-solving and analytical thinking.
- Ability to work in agile delivery teams and collaborate with cross-functional stakeholders.
- Good communication and documentation skills.
- Ability to learn and evaluate fast-evolving GenAI tools, frameworks, and patterns.

Preferred skill sets:

Exposure to Kafka or messaging systems for asynchronous processing.

- Exposure to MCP server development, tool integration patterns, or agent-to-tool communication.
- Basic understanding of A2A communication patterns and multi-agent orchestration.
- Familiarity with Docker, CI/CD, GitHub Actions, Azure DevOps, Jenkins, or equivalent.
- Exposure to fine-tuning, SLMs, vLLM, Ollama, or Hugging Face model deployment.
- Frontend integration awareness using React, Angular, Streamlit, Gradio, or similar.
- Domain exposure in banking, healthcare, insurance, retail, telecom, or enterprise operations.

Job Description & Summary: PwC India is seeking a skilled LLM Observability & Performance Test Engineer to monitor, troubleshoot, and optimize AI-powered applications. This role is crucial for ensuring our large language model systems are performant, reliable, and cost-effective in real-world scenarios by combining robust performance engineering practices with specialized LLM observability techniques.

Responsibilities:

Design and Execute Performance Tests: Develop and implement comprehensive test plans and scripts to evaluate the performance, scalability, and stability of LLM applications under various loads. **Implement LLM Observability:** Instrument LLM applications to capture rich telemetry data, including prompts, responses, token usage, latency, and error information, using specialized tools and frameworks like Datadog, LangChain, or OpenTelemetry. **Monitor and Analyze Metrics:** Track key performance indicators (KPIs) such as response time, throughput, cost per query, accuracy, and resource utilization using real-time dashboards and monitoring systems. **Identify and Mitigate Bottlenecks:** Analyze performance test results and production data to pinpoint performance bottlenecks, errors, and potential issues (e.g., high latency in RAG pipelines) and collaborate with development teams on optimization. **Conduct Automated Evaluations:** Implement automated quality checks and evaluations (e.g., hallucination detection, toxicity classifiers, relevance scoring) to continuously assess model output quality. **Troubleshoot Production Issues:** Utilize tracing and logging data to quickly diagnose the root cause of issues in complex LLM workflows and agentic applications. **Ensure Security and Compliance:** Monitor model behavior for potential security risks, such as prompt injections or sensitive data leaks, and ensure compliance with data protection regulations. **Optimize Costs:** Track and manage token usage and computational resource consumption to ensure cost-effectiveness and alert teams to potential budget overruns. **Collaborate and Report:** Work closely with data scientists, ML engineers, and QA teams to provide actionable insights and recommendations for model fine-tuning and system architecture improvements.

Mandatory skill sets:

Seeking a skilled LLM Observability & Performance Test Engineer to monitor, troubleshoot, and optimize AI-powered applications. This role is crucial for ensuring our large language model systems are performant, reliable, and cost-effective in real-world scenarios by combining robust performance engineering practices with specialized LLM observability techniques.

Strong knowledge of performance testing methodologies and load testing tools such as JMeter, LoadRunner, or Gatling. Familiarity with the unique challenges of LLMs, including non-determinism, hallucinations, and prompt sensitivity. Experience with LLM observability platforms and tools (e.g., Datadog LLM Observability, Arize AI, Langfuse) is highly desirable. Proficiency in programming/scripting languages (e.g., Python, Java). Excellent analytical problem-solving, and communication skills.

Preferred skill sets:

- Relevant certifications in LLM Observability & Performance Test Engineer are a plus

- AppDynamics, DataDog is an added advantage